

PII: S0045-6535(97)00327-5

Photodegradation of Dyes in Aqueous Solutions Containing Fe(III)-oxalato Complexes

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(Received in USA 11 March 1997; accepted 18 June 1997)

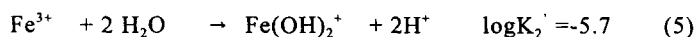
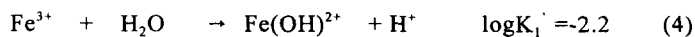
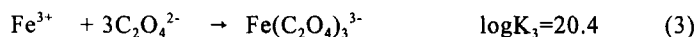
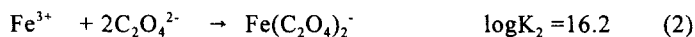
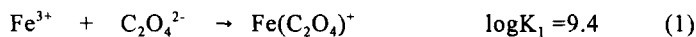
Abstract

The photolysis of Fe(III)-oxalato complexes can produce active oxidant H_2O_2 and $\cdot\text{OH}$ radicals, which can oxidize organic compounds in the atmospheric waters. Fe(III)-oxalato complexes and its photochemical reactions were applied to exploratory research on waste water treatment by promoting the destruction of dye molecules and the discoloration of their solutions. The photodegradation kinetics of the five dyes (C.I. reactive red 2, C.I. reactive blue 4, C.I. reactive black 8, C.I. basic red 13 and C.I. basic yellow 2) in aqueous solutions containing Fe(III)-oxalato complexes were studied. Results showed that the photodegradation reactions of the five dyes were all pseudo-first order reactions. The effects of pH, dye concentration and the ratio Fe(III)/oxalato(Fe/OX) on the photodegradation rate of the dye (C.I. reactive red 2) were also inspected. © 1997 Elsevier Science Ltd

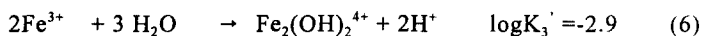
Key Word: Fe(III)-oxalato complexes, UV irradiation, Photodegradation, Dyes,

Introduction

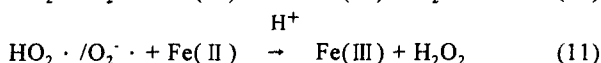
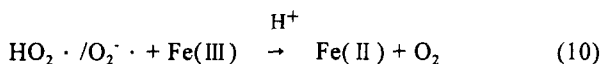
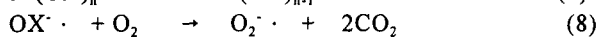
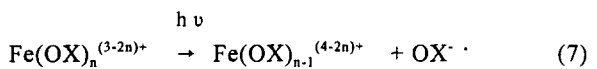
The speciation of dissolved Fe(III) in aqueous solution containing oxalate depends on the competition between the formation of Fe(III)-oxalato complexes and of Fe(III)-hydroxy complexes:



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Calculations show that under the condition of acidic solutions, where the pH range from 3 to 5, Fe(III)-oxalato complexes could be the predominant dissolved species [1]. Since the 1950s, numerous investigations have found that Fe(III)-oxalato complexes photolyze with high efficiency [1,7-11]. Zuo et al. have shown that the photolysis of Fe(III)-oxalato complexes is a major pathway for the formation of H_2O_2 in atmospheric waters [1,2,5]. Iron is also capable of catalyzing the oxidation of dissolved SO_2 and organic substances such as formaldehyde, oxalic and pyruvic acids in cloud droplets [3,6]. The scheme of possible photochemical reactions in the Fe(III)-oxalato-oxygen system was shown by Zuo et al. [1] as below:

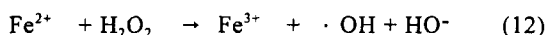


Shown as reaction(7), absorption of a photon by a Fe(III)-oxalato initiates the formation of short-lived intermediates [8,9] that ultimately yield Fe(II) and a free oxalate radical outside of the solvent cage.

Competition between O_2 and Fe(III) species for the oxalate radical ($\text{C}_2\text{O}_4^{2-}\cdot$) and the carbon-centered radical ($\text{C}\cdot$) derived therefrom was noted [9]. If the $\text{C}_2\text{O}_4^{2-}\cdot$ and $\text{C}\cdot$ radicals react with Fe(II), then additional Fe(III) is formed; however, if they react with O_2 , then a sequence of oxidants ($\text{O}_2^\cdot / \text{HO}_2^\cdot$, H_2O_2 , $\cdot\text{OH}$, $\text{ROO}\cdot$ and ROOH) are formed [2].

Zuo et al. [1] detected the H_2O_2 formation in the solutions containing $10\text{ }\mu\text{M}$ Fe(III) and $120\text{ }\mu\text{M}$ oxalate at pH 4 where trioxalate and dioxalate complexes are the predominant iron species, they found the observed rate of formation of H_2O_2 in September noon sunlight was 80 nM s^{-1} . The formation rate of H_2O_2 is related to pH, irradiation intensity and the concentrations of oxalate and dissolved iron.

The H_2O_2 formed in the Fe(III)-oxalato system can react with Fe^{2+} according to a Fenton mechanism [12]:



Zuo [1] indicated that the production of $\cdot\text{OH}$ radicals in atmospheric water droplets has considerable significance as they can oxidize a wide variety of natural and anthropogenic

organic and inorganic substances.

At present, the developing photochemical methods for waste water treatment is increasingly drawing the attentions of those who take part in environmental protection. High-performing catalysts are an important part of the photochemical reaction. Recently, Fe(III)-OH complexes were used as a kind of catalyst in the photodegradation of dyes [13]. In this work, Fe(III)-oxalato complexes are used as the substitute for Fe(III)-OH complexes to promote the photodegradation and discoloration of dyes solutions. The effect of pH and the concentration of Fe(III) and oxalate are investigated. We investigated the photodegradation of dyes solutions by Fe(III)-oxalato complexes to (i) assess the potential utility of photocatalysis for dye solutions; (ii) further our understanding of the photo-reactivity of Fe(III)-oxalato complexes with dyes.

Experimental section

Materials. Dyes and their chemical structures used in this study are the same as those displayed in the previous paper[13]. Dyes were used without further purification. The source of iron was ferric nitrate, and pH of the stock solutions were adjusted to < 1 by adding HNO_3 . Sodium oxalate was used as oxalate source.

All reagents were analytical grade and the water was distilled.

Equipment and Analyses. Irradiation was performed with a UV disinfection lamp ($\lambda_{\text{max}} = 253.7\text{nm}$, 30W). Concentrations of dyes were detected with spectrophotometry UV-120-02 (Shimadzu). pH was measured using a pHs-10B digital pH-meter.

Procedures. The 5 dyes solutions of different concentrations (from 5mg/L to 25mg/L) were prepared in water. In the study on photodegradation kinetics, the same volume solutions of Fe^{3+} and oxalate were added and diluted to a certain volume and the pH was adjusted to 4 with NaOH. The resultant solutions contained $20\ \mu\text{M}$ Fe(III) and $240\ \mu\text{M}$ oxalate and a certain concentration of dyes corresponding to different dyes. The solutions prepared above were irradiated by UV light. The concentrations of the dyes were detected by UV spectrometry at time intervals of 10 minutes.

In order to examine the effect of pH on the photodegradation of dyes in the solutions containing Fe(III)-oxalato complexes, pH2, 3, 4, 5, 6 were chosen as the different pH values. A typical dye (C.I. reactive red 2) was used as an object for treatment. Different photodegradation rates were detected as above and compared with each other in the profile of kinetics constants vis pH.

To investigate the effect of the concentration of dyes on its photodegradation rates, different concentrations of 10mg/L, 20mg/L and 40mg/L were used as the initial concentrations of the dye (C.I. reactive red 2). To exam the effect of the ratio Fe^{3+} :oxalate (Fe/OX) on the photodegradation rates of the dye, the dye solutions containing different concentrations of Fe^{3+} and oxalate were used in the experiments.

Results and Discussion

1. The effect of photolysis of Fe^{3+} -oxalate complexes on the discoloration of the dyes solutions

Shown as the previous paper[13], the photolysis of Fe^{3+} -OH complexes ($\text{Fe}(\text{OH})^{2+}$) can promote the degradation of the dyes molecules and the discoloration of their solutions at pH 3. The species responsible for the reactions is believed to be hydroxyl radicals formed by the photochemical dissociation of Fe^{3+} -hydroxy complexes. The results suggested that the chromophere of the dyes in the visible region was distroied by $\cdot\text{OH}$ radicals.

In order to distinguish the effect of Fe^{3+} -OH complexes, the control experiment was conducted, the result of which was shown in Figure 1.

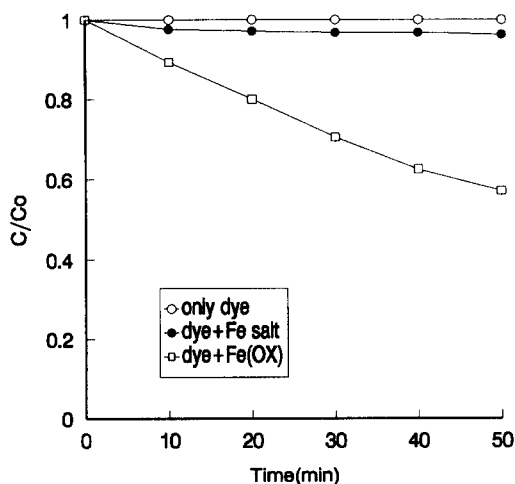


Figure 1. The control experiment of photodegradation of the dye C.I. reactive red 2 in the solutions at pH 4 respectively containing Fe^{3+} , $\text{C}_2\text{O}_4^{2-}$, and Fe^{3+} -oxalato complexes. [Dye]=20mg/L, $[\text{Fe}(\text{III})]=20\ \mu\text{M}$, $[\text{C}_2\text{O}_4^{2-}]=240\ \mu\text{M}$.

Under the condition without $\text{C}_2\text{O}_4^{2-}$ ions, the solution contained only Fe^{3+} -OH complexes, and the discoloration of the dye solution is less obvious than that of the dye solution containing

$\text{Fe}^{3+}\text{-C}_2\text{O}_4^{2-}$ complexes, which indicated that ,under the same conditions(pH, CFe^{3+}), the photolysis of the later has a greater capability of promoting the discoloration of the dye solutions than that of the former.

Additionally, there was a negligible discoloration of the dye solutions without Fe^{3+} ions, and the self-photodegradation of the dye could also be neglected.

From above, a conclusion can be made that the active oxidants species H_2O_2 and $\cdot\text{OH}$ formed in the photolysis of Fe(III)-oxalato complexes in the aqueous solutions can ,to a certain extent, destroy the dye molecules and cause the discoloration of its solutions, which implied that Fe(III)-oxalato complexes can be used to promote photodegradation of dyes in the waste water treatment.

Furthermore, as Fe(III)-oxalato complexes absorb even more strongly than Fe(III)-hydroxy complexes in the tropospheric solar UV-visible region (290-570nm) and are also photochemically more reactive[1], the Fe(III)-oxalato complexes system has the potential of utilizing sunlight as irradiation source to promote the degradation of organic compounds. Moreover, as one of the most abundant elements and its extensive distribution , Fe(III) species will be a cheap catalyst.

2. Photodegradation kinetics of dyes in aqueous solutions containing Fe^{3+} -oxalato complexes

The method used in study on photodegradation kinetics of the five dyes was the same as described in the previous paper [13]. The results gained were shown in Table 1 and Figure 4.

Table 1. Photodegradation kinetics data

($n=6$, $r_c=0.9249$)

Dye	$-r_1$	reaction order	linear regression equation	k (min^{-1})
C.I.reactive blue 4	0.9301	first	$\ln C/C_0 = -0.0162 - 0.0010T$	0.0010
C.I.reactive black 8	0.9950	first	$\ln C/C_0 = -0.0020 - 0.0054T$	0.0054
C.I.reactive red 2	0.9992	first	$\ln C/C_0 = -0.0114T$	0.0114
C.I.basic yellow 2	0.9911	first	$\ln C/C_0 = 0.0020 - 0.0154T$	0.0154
C.I.basic red 13	0.9897	first	$\ln C/C_0 = -0.1305 - 0.0407T$	0.0407

r_c : critical correlative coefficient ($\alpha=0.001$)

r_1 : calculated correlative coefficient of pseudo-first order reaction

k: kinetics constant

T: time of the irradiation in the unit of minute

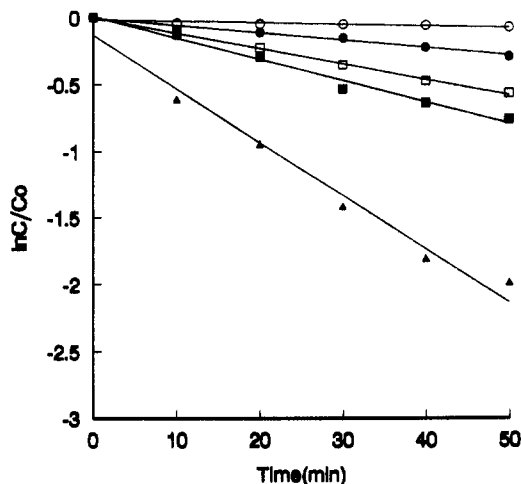


Figure 2. Photodegradation rates of dyes in the solutions at pH4 containing $20 \mu\text{M}$ Fe(III) and $240 \mu\text{M}$ $\text{C}_2\text{O}_4^{2-}$.

(○) C.I. reactive blue 4. $C_0 = 25\text{mg/L}$

(●) C.I. reactive black 8. $C_0 = 20\text{mg/L}$

(□) C.I. reactive red 2. $C_0 = 20\text{mg/L}$

(■) C.I. basic yellow 2. $C_0 = 5\text{mg/L}$

(▲) C.I. basic red 13. $C_0 = 8\text{mg/L}$

The photodegradation of the five dyes were all pseudo-first order reactions. With different photodegradation rates, the discoloration of the five dyes solutions had taken place, which could be detected by spectrophotometry. Especially, for C.I. basic red 13, the photodegradation reaction was nearly 1 order of magnitude faster than that of the same dye in the solution at pH3 containing 20mg/L Fe(III) under the irradiation of the same UV light investigated in the previous paper[13]. For different dyes, photodegradation rates are of difference, which is related to the degradability of dye molecules.

With less addition of Fe(III) and comparatively higher photolysis rate, Fe(III)-oxalato complexes could be used as a better photocatalyst in photochemical treatment for waste water of organic compounds.

3. The effect of pH on the photodegradation rates of the dye in the solutions containing Fe^{3+} -oxalato complexes.

The competition between formation of Fe(III)-oxalato complexes and Fe(III)-hydroxy complexes has an important effect on the photoformation of H_2O_2 and $\cdot\text{OH}$ radicals in aqueous solutions containing Fe(III) and oxalate. The pH and concentration ratio of Fe(III) to oxalate are two of the most important elements and parameters which to some extent determine the species distribution of iron(III) species.

The pH effects on the photolysis reaction in the Fe(III)-oxalato complexes-dissolved oxygen system primarily was embodied as the determination for Fe(III) species contribution (i.e. which is the predominant species, Fe(III)-oxalato complexes or Fe(III)-hydroxy complexes). On the other hand, pH affects the competition between the reactions of active intermediates $\text{HO}_2\cdot / \text{O}_2^{\cdot-}$ and Fe(III)/Fe(II) [1] (reaction (10), (11)), which thus affects the formation rate of H_2O_2 and $\cdot\text{OH}$ radicals. In fact, these two aspects are integrated into one point that pH has an important effect on the photochemical efficiency of the system.

Zuo et al.[1] pointed out that in the Fe(III)-oxalato solution at low pH range 1.5-4.0, the H_2O_2 photoformation rate stayed at a rather high and constant value. Above pH 4.0, the rate of H_2O_2 formation decreased with increasing pH, under which condition the O_2 is the main product rather than H_2O_2 . In addition, above pH 5.5, the fraction of Fe(III)-oxalato complexes decreases sharply, which further reduces the possibility of H_2O_2 formation.

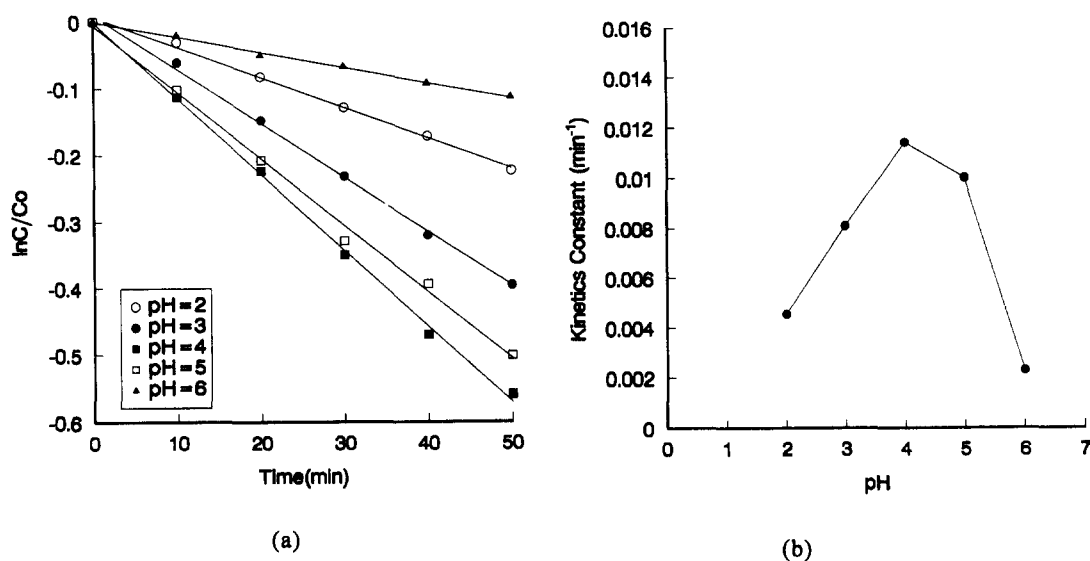


Figure 3. The pH effect on photodegradation rate.

Figure 3 showed that ,at pH3-4 , the photodegradation kinetics constants remained at high values. At other pH range the kinetics constants were smaller , especially at higher pH range ($\text{pH} > 5$), the kinetics constants reduced sharply with increasing pH. This result was in a good agreement with the result gained by Zuo and his coworkers[1].

Thus , pH4 was rationally chosen as the constant pH index for photodegradation of all the dyes in this work.

4. Dye concentration effect on its photodegradation rates in the aqueous solution containing Fe(III)-oxalato complexes.

To examine the effect of the initial concentration of the dye on its photodegradation rates, the solutions of different dye concentration of 10-40mg/L containing Fe(III)-oxalato complexes ($\text{Fe/OX} = 20 \mu\text{M}/240 \mu\text{M}$) were irradiated by UV light. The photodegradation rate decreased with increasing its initial concentration in the range 10-40mg/L (see Figure 4). In the lower concentration range ($< 40\text{mg/L}$), the photodegradation of the dye was pseudo-first order reaction.

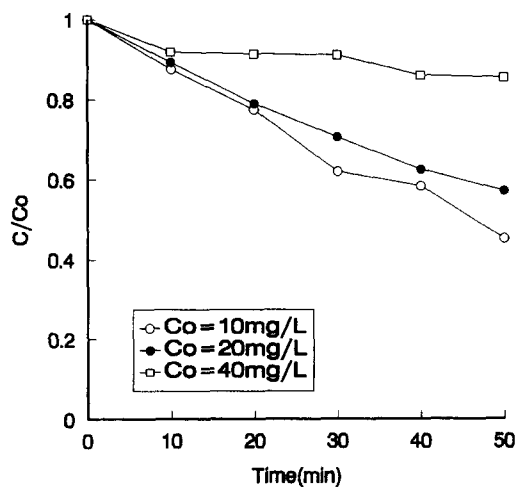


Figure 4. Dye concentration effect on photodegradation rate of the dye in the solution at pH4 containing Fe(III)-oxalat complexes.

5. The effect of the ratio of Fe^{3+} to oxalate (Fe/OX) on the photodegradation of the dye solutions

At a given pH, the concentration ratio of Fe(III) to oxalate can determine the species

At a given pH, the concentration ratio of Fe(III) to oxalate can determine the species distribution of Fe(III)-oxalato complexes in the aqueous solutions. As the ratio (Fe/OX) was always at least 1:10, the only significant iron species is the trioxalate complex [7].

To determine the dependence of photodegradation rate on the Fe(III) concentration, the dye (C.I. reactive red 2) solutions at pH4 containing 10-40 μ M Fe(III), 240 μ M oxalate were irradiated. The photodegradation rate increased with increasing Fe(III) concentration over the range 10 -40 μ M (see Figure 5). The result was in accordance with speciation calculation which showed that the total concentration of ferric-oxalato complexes increased linearly with Fe(III) under the above conditions.

To test the oxalate concentration effect, Fe(III) was held at 20 μ M and the total oxalate varied from 120 μ M to 480 μ M. The result, presented in Figure 6, indicated that the photodegradation rate increased with increasing oxalate concentration in the range 120-480 μ M.

The Fe(III) and oxalate concentrations affect the photodegradation rate more significantly in 10-20 and 120-240 μ M ranges of Fe(III) and oxalate, respectively. The further increase in concentrations of the above components (by a factor of two) results only in a small acceleration of the reaction. These two results are in agreement with that gained by Zuo et al.[1].

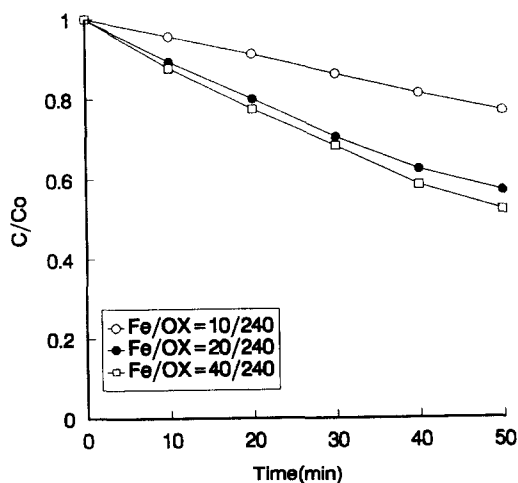


Figure 5. Fe(III) concentration effect on photodegradation rate.

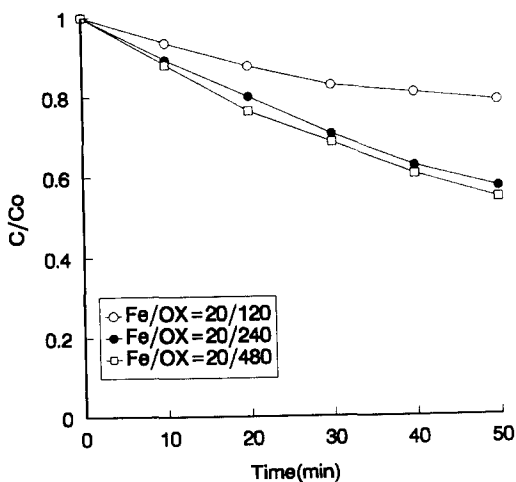


Figure 6. Oxalate concentration effect on photodegradation rate.

Acknowledgments

Sincerely, the authors would like to thank Dr. Y. Zuo (Southeast Environmental Research Program, Florid International University) for his good advice , generous help and enthusiastically discussing the topic with us .

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